

Evaluation of Transfer Learning Methods Applied on 2D Robotic Arms

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M2 PAR — Master 2nd Year, Artificial Perception and Robotics

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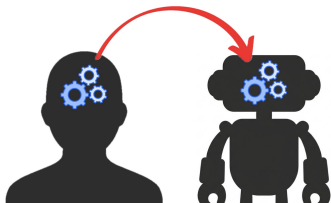
- Romuald Aufrère (jury)
- Sébastien Lengagne (co-supervisor)
- Antonio Claudio de Sousa dos Santos Filho (co-supervisor)

Outline

- 1 Introduction
- 2 State of the art
- 3 Method
- 4 Results
- 5 Conclusion

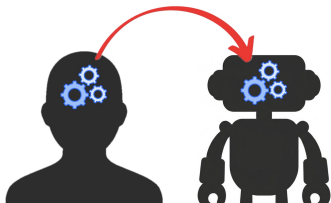
Introduction

Internship:
Human-to-robot transfer

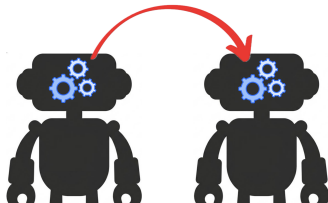


Introduction

Internship:
Human-to-robot transfer



Project:
Robot-to-robot transfer



Objective

Evaluate transfer methods by applying to 2D robots

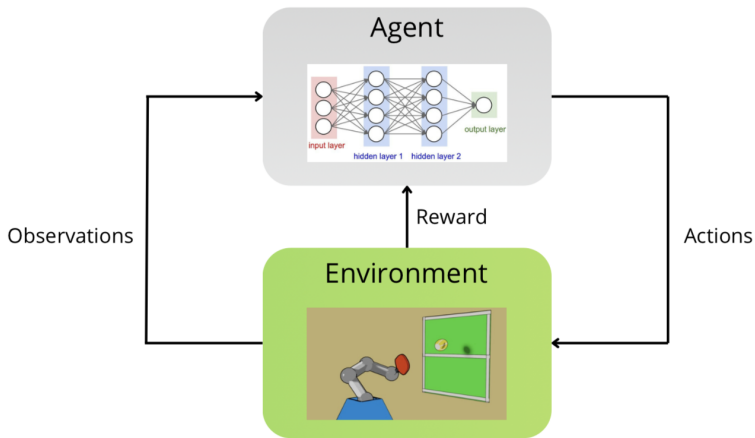
Objective

Evaluate transfer methods by applying to 2D robots

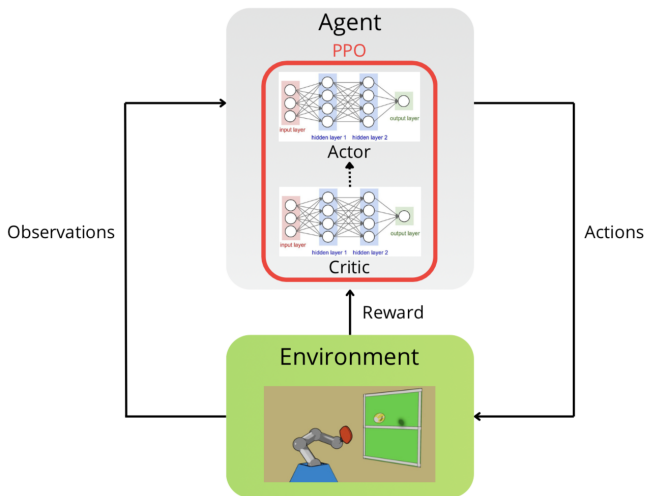
Tools:

- Reinforcement learning
- PPO, Gymnasium, Stable-Baselines3

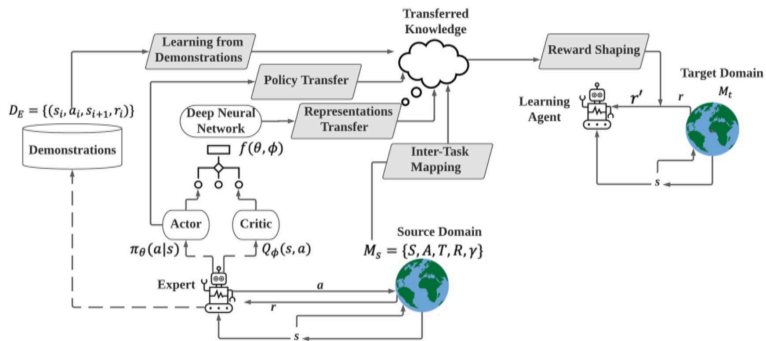
Reinforcement Learning



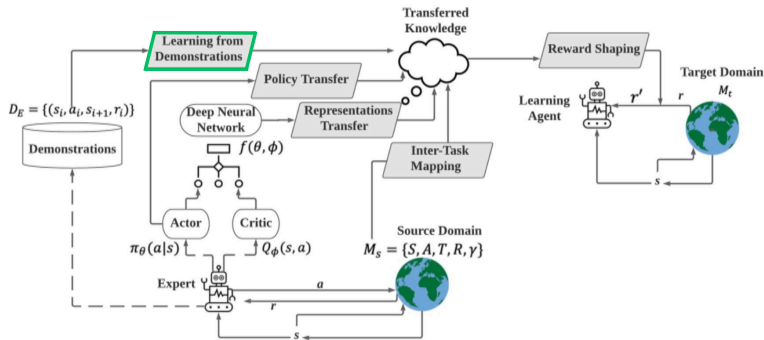
Proximal Policy Optimization



State of the art

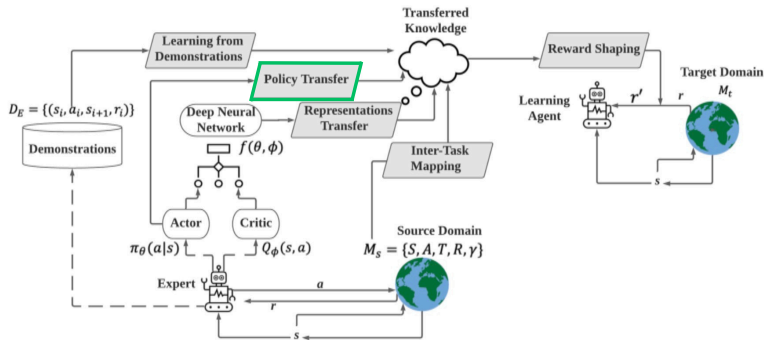


State of the art



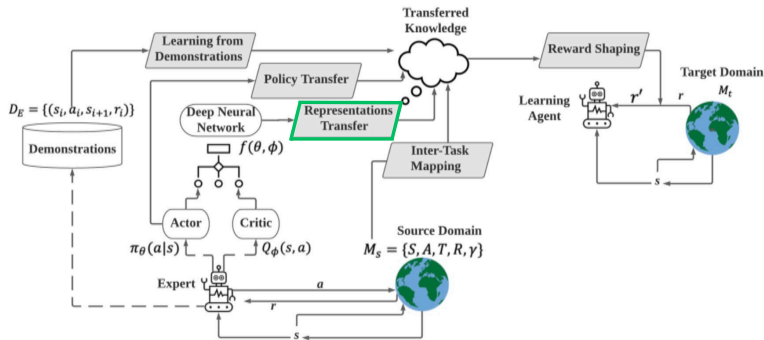
Learning from Demonstration: Accelerates learning by leveraging expert trajectories to guide exploration and policy initialization

State of the art



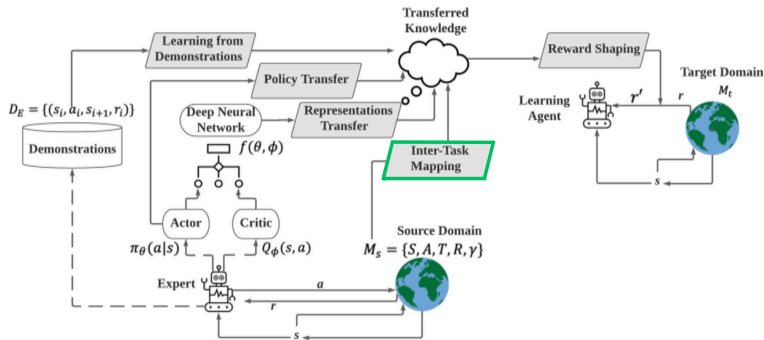
Policy Transfer: Direct reuse or fine-tuning of pre-trained source parameters to speed up target task convergence

State of the art



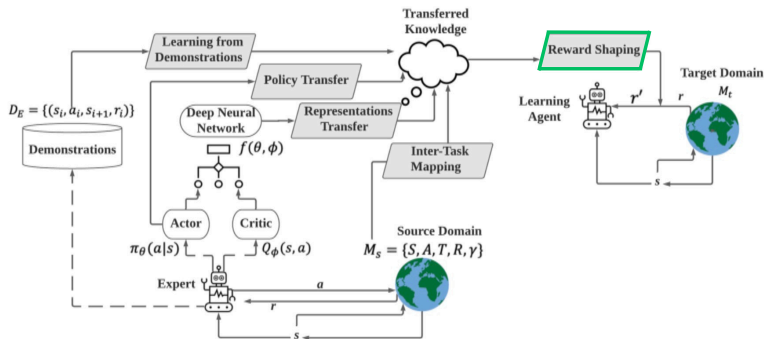
Representation Transfer: Transfers learned features or latent state models to handle complex observations in the target domain

State of the art



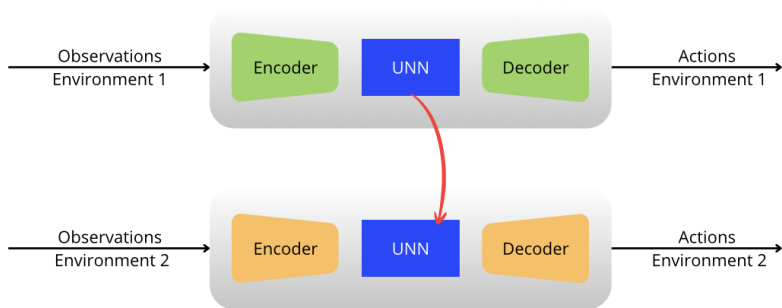
Inter-Task Mapping: Employs mapping functions to translate states and actions between structurally different source and target domains

State of the art

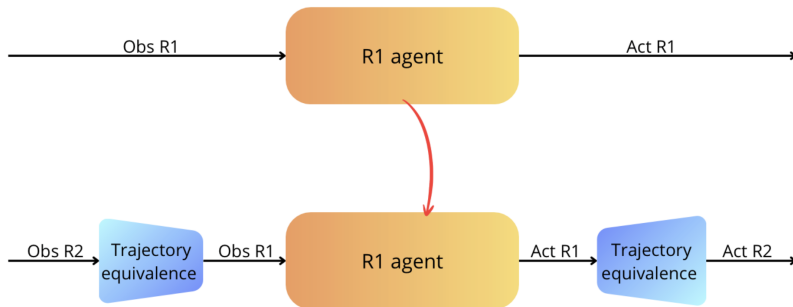


Reward Shaping: Uses source knowledge to design auxiliary rewards, providing denser feedback for faster optimization

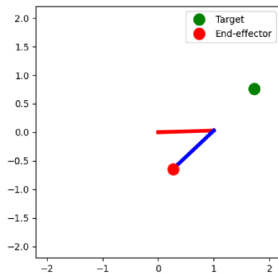
Universal Notice Network



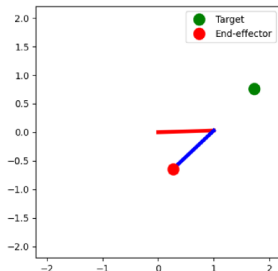
Transfer Method



2-DoF Arm — Reaching Task



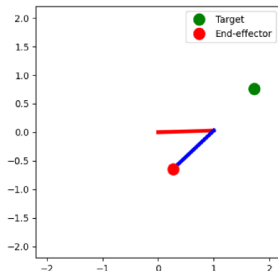
2-DoF Arm — Reaching Task



Observations:

- Joint 1 angle
- Joint 2 angle
- Distance effector/target on x
- Distance effector/target on y

2-DoF Arm — Reaching Task



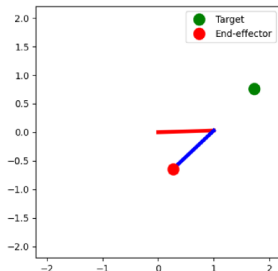
Observations:

- Joint 1 angle
- Joint 2 angle
- Distance effector/target on x
- Distance effector/target on y

Actions:

- Angular velocity joint 1
- Angular velocity joint 2

2-DoF Arm — Reaching Task



Observations:

- Joint 1 angle
- Joint 2 angle
- Distance effector/target on x
- Distance effector/target on y

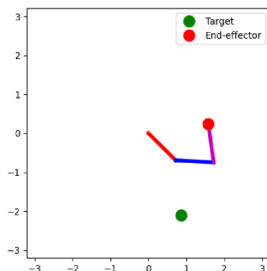
Actions:

- Angular velocity joint 1
- Angular velocity joint 2

Reward:

- $-Distance_{effector / target}$

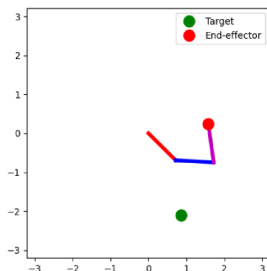
3-DoF Arm — Reaching Task



Observations:

- Joint 1 angle
- Joint 2 angle
- Joint 3 angle
- Distance effector/target on x
- Distance effector/target on y

3-DoF Arm — Reaching Task



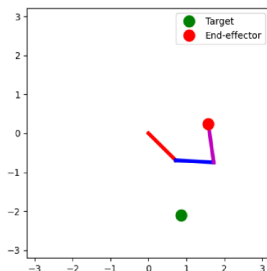
Observations:

- Joint 1 angle
- Joint 2 angle
- Joint 3 angle
- Distance effector/target on x
- Distance effector/target on y

Actions:

- Angular velocity joint 1
- Angular velocity joint 2
- Angular velocity joint 3

3-DoF Arm — Reaching Task



Observations:

- Joint 1 angle
- Joint 2 angle
- Joint 3 angle
- Distance effector/target on x
- Distance effector/target on y

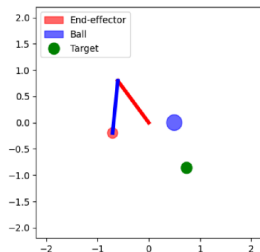
Actions:

- Angular velocity joint 1
- Angular velocity joint 2
- Angular velocity joint 3

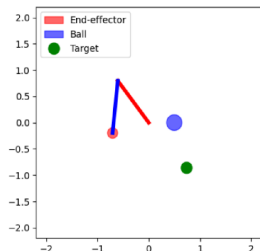
Reward:

- $-\text{Distance}_{\text{effector} / \text{target}}$

2-DoF Arm — Push Ball Task



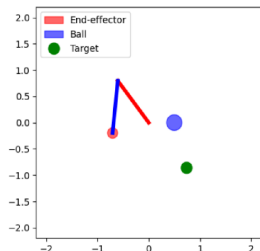
2-DoF Arm — Push Ball Task



Observations:

- Joint 1 angle
- Joint 2 angle
- Distance effector/ball (x, y)
- Distance ball/target (x, y)

2-DoF Arm — Push Ball Task



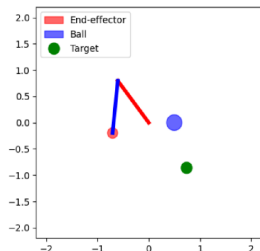
Observations:

- Joint 1 angle
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- Distance effector/ball (x, y)
- Distance ball/target (x, y)

Actions:

- Angular velocity joint 1
- Angular velocity joint 2

2-DoF Arm — Push Ball Task



Observations:

- Joint 1 angle
- Joint 2 angle
- Distance effector/ball (x, y)
- Distance ball/target (x, y)

Actions:

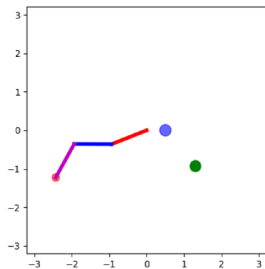
- Angular velocity joint 1
- Angular velocity joint 2

Reward:

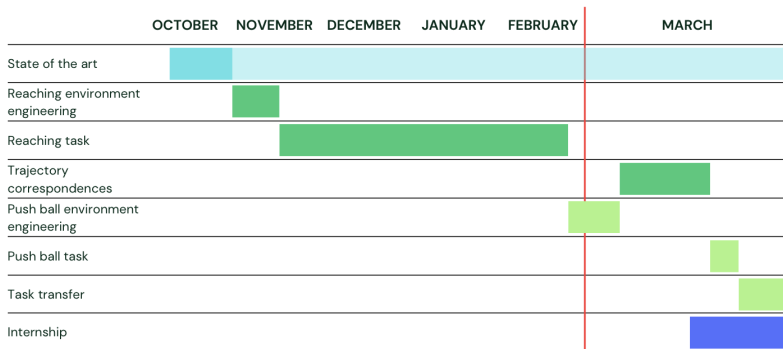
- $-\text{Distance}_{\text{ball}/\text{target}}$

3-DoF Arm — Task Transfer

- Use previously learned trajectories
- Transfer Push Ball task on 3 DoF arm



Project Planning — Gantt Summary



Conclusion

- Better understanding of RL, tools and methods
- Robot-to-robot transfer
- Solid basis for the upcoming research internship :
Human-to-robot transfer

Thank You

Thank you for your attention!

Any questions?